



8/7/2026

# ASSIGNMENT DAY2

Router & Switch Fundamentals.

## TASK 1.

- ❖ Draw a labelled diagram of a typical home Wi Fi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

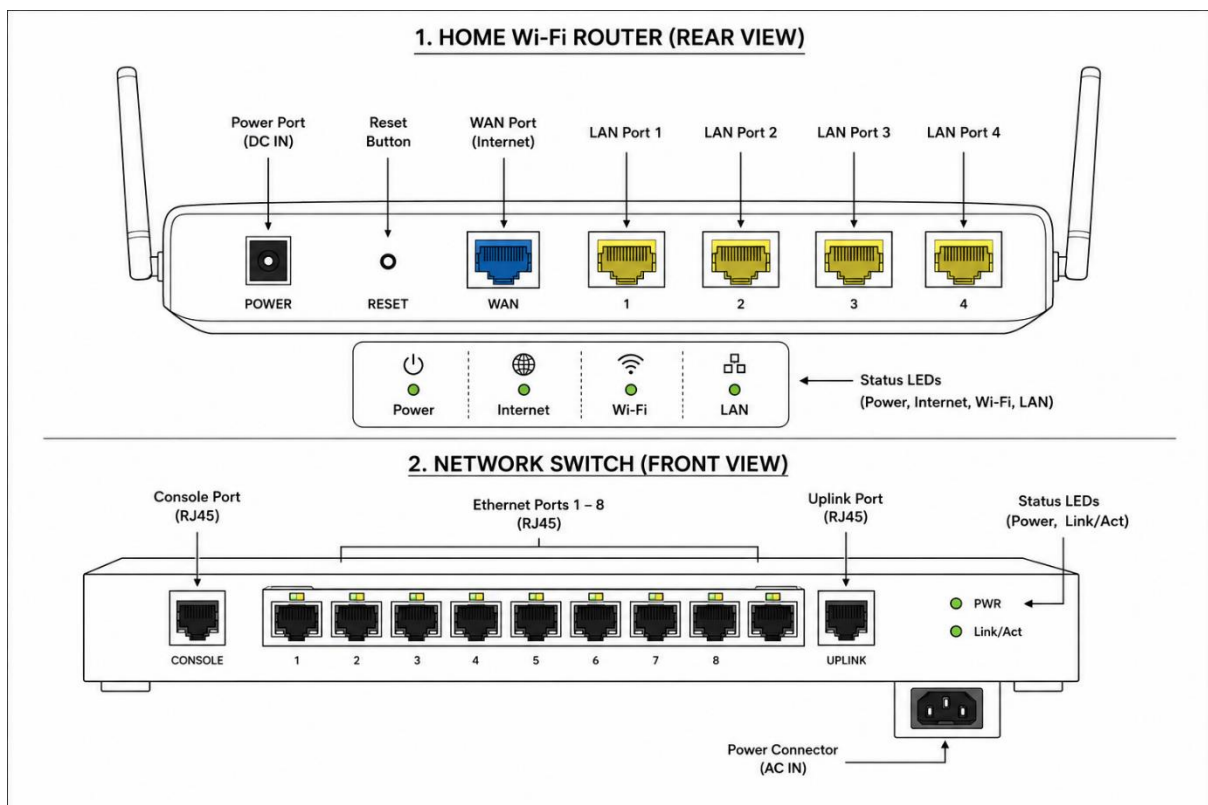


Figure : Labelled Diagram of a Home Wi Fi Router (Rear View) and a Network Switch (Front View).

### Explanation :

- ✓ The diagram shows the main parts of a home Wi-Fi router and a network switch. It clearly labels the WAN port, LAN ports, Console port, Power Connector, and the Status LEDs. These ports are used to connect network devices, while the LEDs show the power and network connection status.

### Summary :

- ✓ This Labelled diagram helps identify the important ports and status LEDs of a router and a network switch. It makes it easier to understand their basic connections and functions in a computer network.

## TASK 2.

- ❖ **List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

### ➤ Key Differences Between Routers and Switches :

| <u>Feature</u>         | <u>Router</u>                                                        | <u>Switch</u>                                                                  |
|------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Main Function</b>   | Connects different networks, such as a home network to the Internet. | Connects multiple devices within the same local network (LAN).                 |
| <b>Address Used</b>    | Uses IP addresses to send data between different networks.           | Uses MAC addresses to send data to the correct device within the same network. |
| <b>Network Traffic</b> | Routes data between different networks and chooses the best path.    | Forwards data only within the local network, making communication faster.      |

### TASK 3.

- ❖ Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

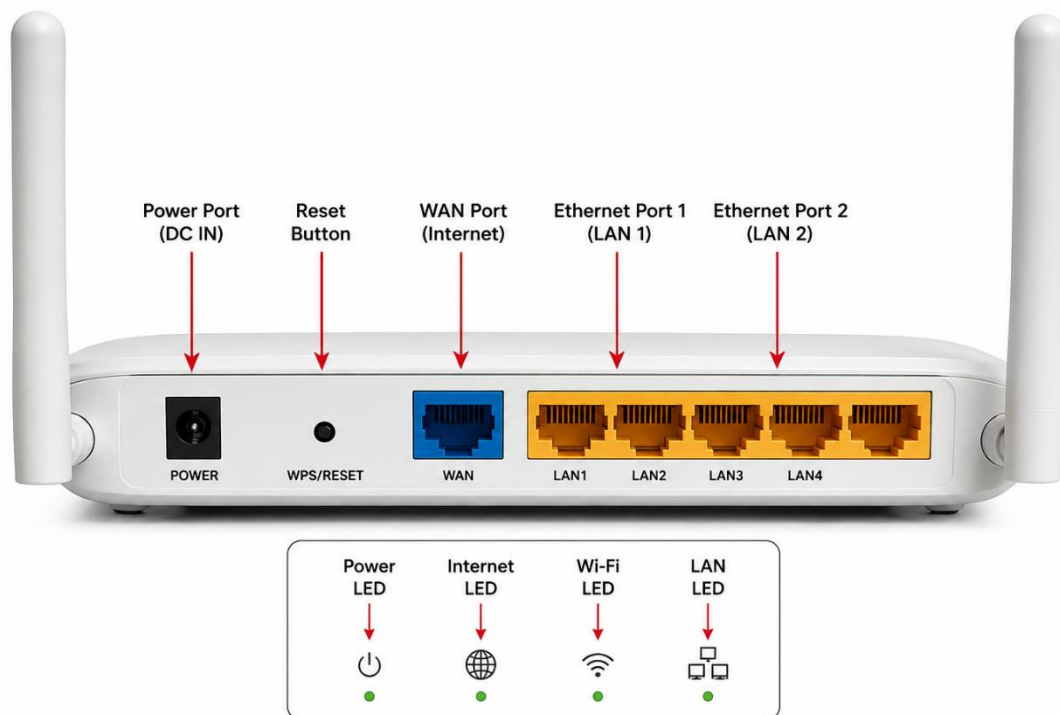


Figure : Real Home WI FI Router (Rear View) with Labelled Ports and LED Indicators.

#### Explanation :

- ✓ The image shows the rear view of a home Wi-Fi router. The Power Port (DC IN) is used to connect the power adapter. Ethernet Port 1 (LAN 1) and Ethernet Port 2 (LAN 2) are used to connect devices such as a computer or another network device using an Ethernet cable. The LED

indicators (Power, Internet, Wi-Fi, and LAN) display the router's power status and network activity.

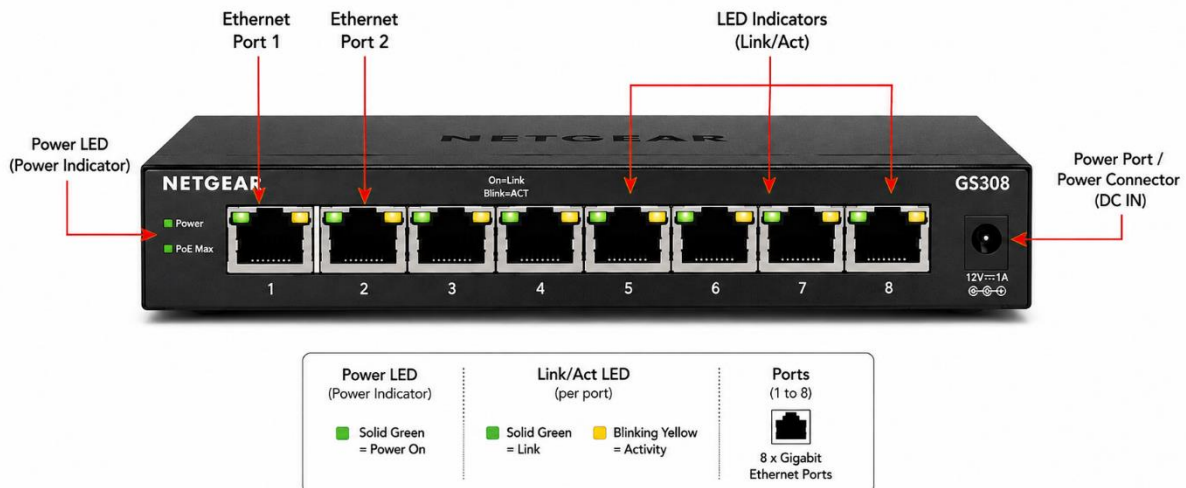


Figure 2: Real Network Switch (Front View) with Labelled Ports and LED Indicators.

*Figure: Real Network Switch (Front View) with Labelled Ports and LED Indicators.*

### Explanation :

- ✓ The image shows the front view of a network switch. The Power Port (DC IN) supplies power to the switch. Ethernet Port 1 and Ethernet Port 2 are used to connect computers and other network devices. The LED indicators (Power and Link/Act) show whether the switch is powered on and if the connected ports are active.

## **TASK 4.**

❖ **Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.**

### **➤ Comparison Between Router and Switch :**

#### **Scenario :**

- ✓ I am setting up a small gaming café for IPL streaming and multiplayer gaming. First, I will connect the router to the ISP line because it provides Internet access. Next, I will connect the router to a network switch. Then, I will connect all 8 gaming PCs to the switch using Ethernet cables for a fast and stable network.

#### **Justification :**

- ✓ The router connects the café to the Internet and manages traffic between the ISP and the local network. The switch connects all 8 gaming PCs, allowing fast communication and smooth performance for online games and IPL streaming.